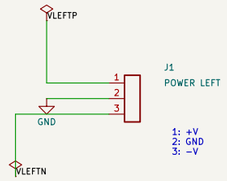
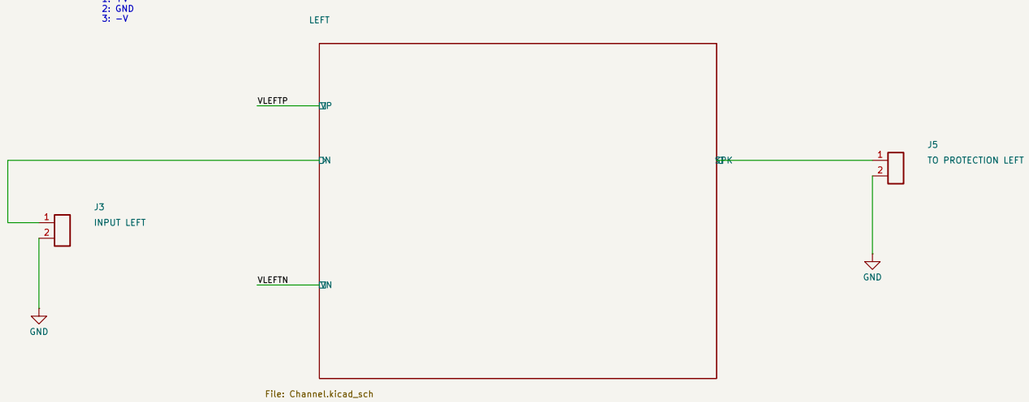
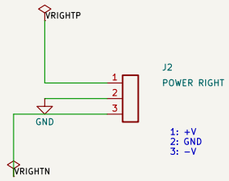
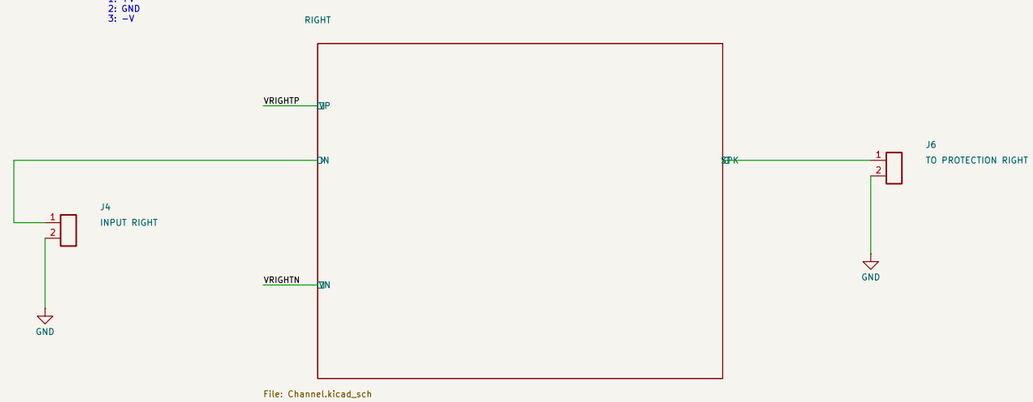
MOSFETAmp2 — STEREO DISCRETE CLASS-AB

REV 0.1: SIMULATION DEVELOPMENT / NOT FABRICATION RELEASED
8 ohm prototype: external current-limited +/-12 to +/-15 V first.
No mains circuitry. External DC speaker protection required. Read README and bench procedure.
BIAS THERMAL CORNER FAILED: actual-device characterization or revised compensation required.









Assembly gates
1. Fit bias short J101/J201; set trimmers for MAXIMUM resistance.
2. Verify live tab insulation and Q108/Q208 thermal contact.
3. Connect supply GND to the junction of the two series bench outputs.
4. Connect oscilloscope earth clips to board GND only.
5. Use dummy loads; no speakers until DC protection and warm-up tests pass.



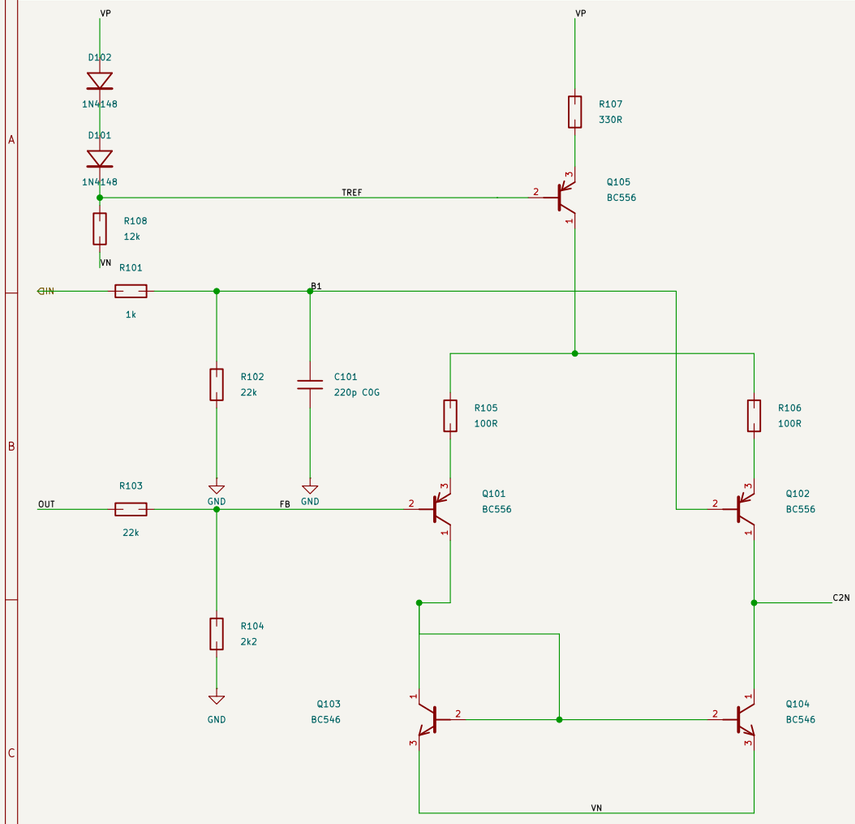
External PSU assumption envelope: roughly +/-12.5 V valley to +/-20.6 V peak.
Final transformer and capacitors not fixed. Recalculate and measure before use.
Schematic connected and ERC checked separately from electrical/model/thermal validation.
The original empty PCB is retained. Layout is for the user; no routing supplied.

Sheet: /		
File: MOSFETAmp2.kicad_sch		
Title: MOSFETAmp2 stereo Class-AB		
Size: A3	Date: 2026-10-25	**Rev: 0.1 EXPERIMENTAL**
KiCad E.D.A. 10.0.5		Id: 1/3

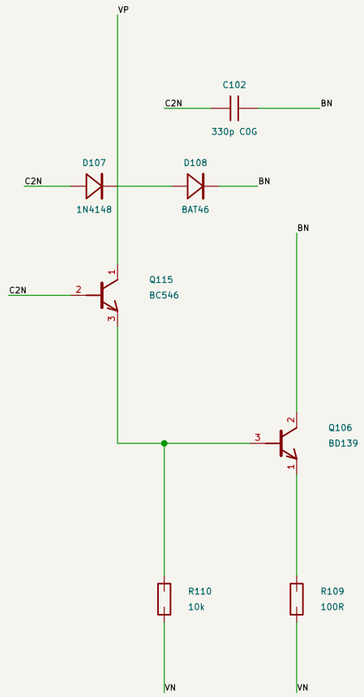
INPUT / DIFFERENTIAL PAIR + MIRROR

BUFFERED VAS / DOMINANT POLE

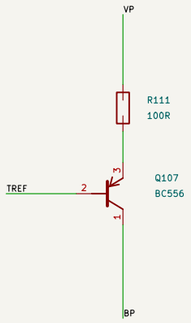
DRIVERS / ONE COMPLEMENTARY PAIR

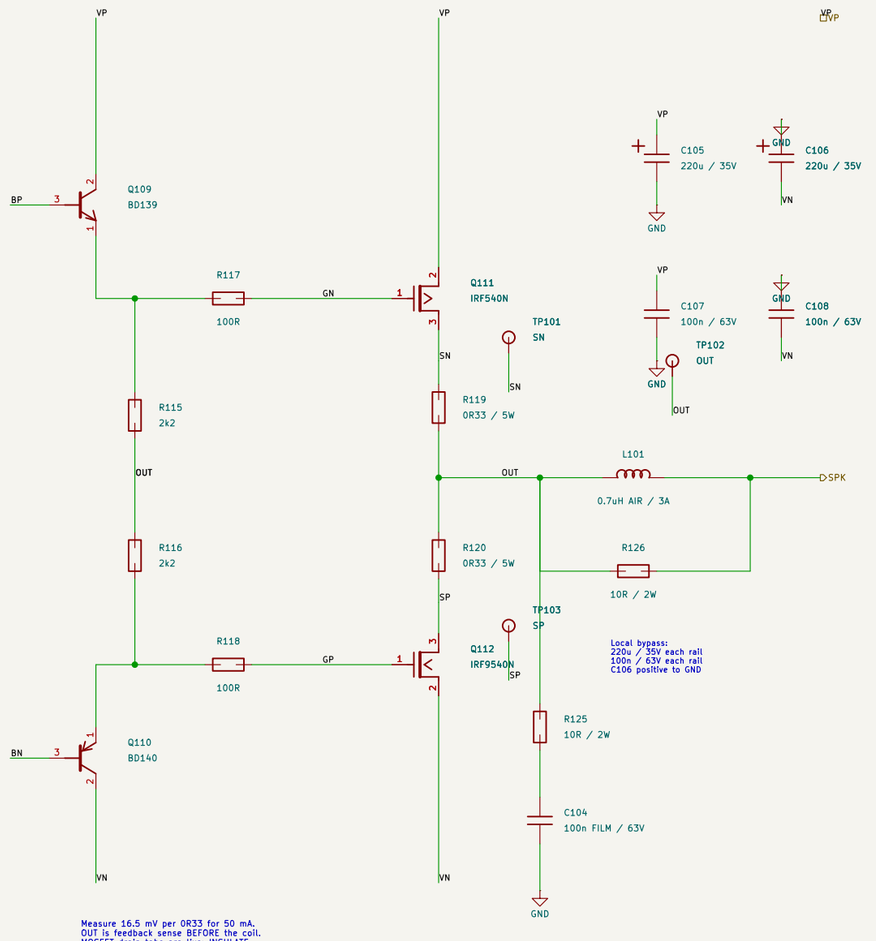


Q101/Q102: match at 1 mA; bond flat faces.
R101 + R102: input about 23k.
Gain = 11 from B1: about 10.52 at IN.
D105/D106 limit differential overload.

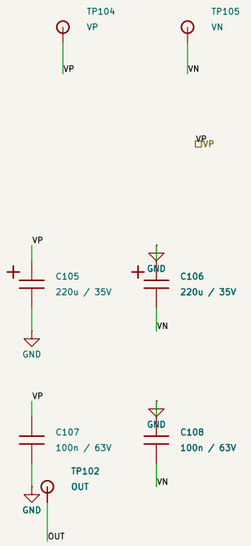


C102 = 330p C0G, 50V, radial 2.54mm.
D107 + D108 prevent VAS saturation.
No rail bootstrapping or regulators.
VAS/load current about 6 mA.





Measure 16.5 mV per 0R33 for 50 mA.
OUT is feedback sense BEFORE the coil.
MOSFET drain tabs are live: INSULATE.
8 ohm prototype; no short-circuit protection.

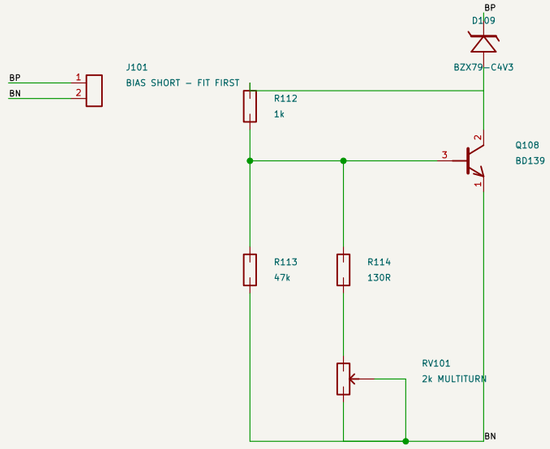


Local bypass:
220u / 35V each rail
100n / 63V each rail
C106 positive to GND

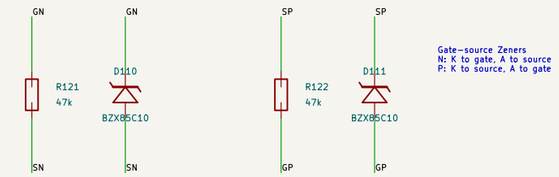
OPERATING LIMITS

8 ohm prototype: current-limited bench PSU first.
No output short-circuit protection is fitted.
External DC speaker protection is required.
Gate clamps protect VGS; they do not establish SOA.
Read the bias and warm-up procedure before power-up.

THERMAL BIAS / FAIL-LOW ADJUSTMENT



Q108 on heatsink beside MOSFETs.
R114 is a bias-range tuning part.
Start RV101 at maximum resistance.
Loss of wiper contact reduces bias.
LOW-VTH HOT CORNER FAILS; see DESIGN.md.



Gate-source Zeners
N: K to gate, A to source
P: K to source, A to gate

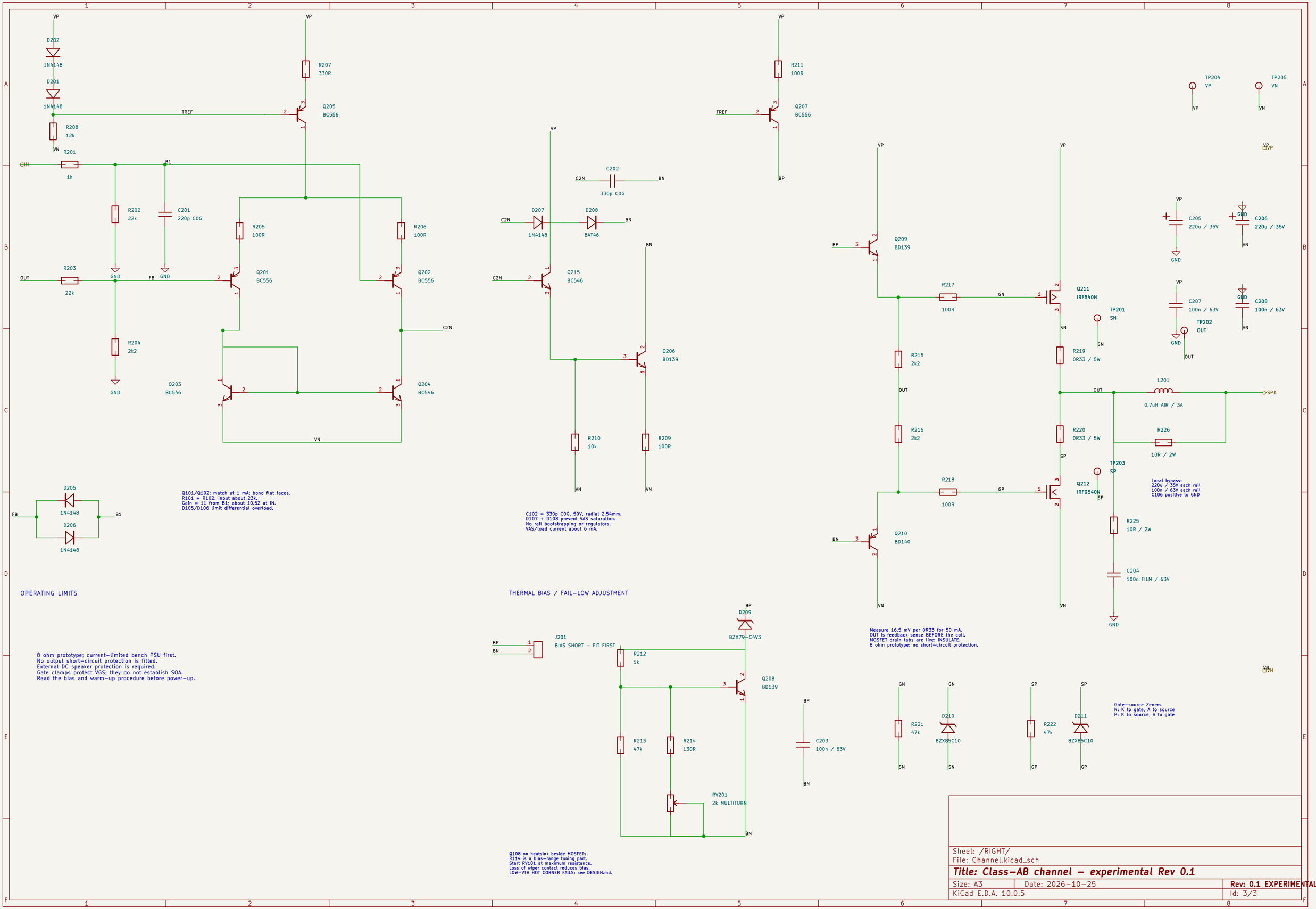
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Title: Class-AB channel - experimental Rev 0.1	
Size: A3	Date: 2026-10-25
KiCad E.D.A. 10.0.5	Id: 2/3

Rev: 0.1 EXPERIMENTAL

INPUT / DIFFERENTIAL PAIR + MIRROR

BUFFERED VAS / DOMINANT POLE

DRIVERS / ONE COMPLEMENTARY PAIR



Sheet: /RIGHT/
File: Channel.kicad_sch

Title: Class-AB channel - experimental Rev 0.1

Size: A3 | Date: 2026-10-25 | **Rev: 0.1 EXPERIMENTAL**

KiCad E.D.A. 10.0.5 | Id: 3/3